

Tristan Bilot

📍 Paris, FR | ✉ Email | 🌐 Website | 🔗 LinkedIn | 🐙 GitHub | 🎓 Google Scholar

Education

Ph.D. in Computer Science , Université Paris-Saclay , Paris, France	2022 – 2025
- Thesis: Detecting Advanced Cyberattacks with Self-Supervised Graph Learning - Supervisors: Pr Khaldoun Al Agha, Dr Nour El Madhoun	
Master's (Diplôme d'Ingénieur) in Computer Science , EPITA , Paris, France	2019 – 2022
- GPA: 4.0/4.0 - Focus on Computer Security, Systems, Network	
DUT in Computer Science , Université Paris-Cité , Paris, France	2017 – 2019
Focus on Algorithms, Data Structures, Data Mining, Reflective Programming	

Experience

Postdoctoral Fellow – University of British Columbia , Vancouver, BC, Canada	Feb. 2026 –
- Research on foundation models for security - Agentic workflows for security	
Applied Scientist Intern – Amazon , New York, NY, USA	Oct. 2025 – Jan. 2026
- Security Analytics and Artificial Intelligence Research team (SAAR) at AWS - Research on multi-agent system security - NeurIPS submission in review	
Ph.D. Researcher – Iriguard , LISN , LISITE , Paris, France	Oct. 2022 – Oct. 2025
- Developed scalable intrusion detection systems with deep learning on client data, focus on APTs and zero-days - Ph.D. funded by Iriguard and in collaboration with LISN and LISITE labs	
Visiting Research Student – University of British Columbia , Vancouver, BC, Canada	Apr. – Jun. 2024
- Research in provenance-based intrusion detection systems with GNNs and self-supervised learning, supervised by Thomas Pasquier - Worked on large-scale temporal graphs and robustness to adversarial attacks	
Student Researcher – EPITA Systems Laboratory (LSE) , Paris, France	Sep. 2021 – Aug. 2022
Research on GNNs for phishing web page detection, supervised by Dr. Badis Hammi	
Data Engineer Apprentice – Carrefour-Google AI Lab , Paris, France	May 2021 – Aug. 2022
- Deployed ML models in production + optimized training time (4x and 3x faster) - Built a scalable BigQuery fetching tool, presented in internal engineering reviews - Deployed a multi-project data pipeline with Airflow, dbt, GCP, Kubernetes	
Software Engineer Apprentice – Carrefour , Paris/Massy, France	Sep. 2019 – Apr. 2021
- Developed new features for the Carrefour iOS app (1.5M+ monthly users) - Integrated Apple Wallet into the app	
Software Engineer Intern – Micropole , Levallois-Perret, France	May 2019 – Aug. 2019
- Developed backend features for websites and web services - Improved website loading speed by ~30%	

Publications

Full list: [Google Scholar](#) – AR: Acceptance Rate

PIDSMaker: Building and Evaluating Provenance-based Intrusion Detection Systems

KDD 2026 (AR: 21%)

[[paper](#), [code](#), [poster](#)]

[Tristan Bilot](#), Baoxiang Jiang, Thomas Pasquier

Sometimes Simpler is Better: A Comprehensive Analysis of State-of-the-Art Provenance-Based Intrusion Detection Systems

USENIX Security 2025 (AR: 17.1%)

[[paper](#), [code](#), [poster](#), [slides](#), [video](#)]

[Tristan Bilot](#), Baoxiang Jiang, Zefeng Li, Nour El Madhoun, Khaldoun Al Agha, Anis Zouaoui, Thomas Pasquier

ORTHRUS: Achieving High Quality of Attribution in Provenance-based Intrusion Detection Systems

USENIX Security 2025 (AR: 17.1%)

[[paper](#), [code](#), [slides](#), [video](#)]

Baoxiang Jiang, [Tristan Bilot](#), Nour El Madhoun, Khaldoun Al Agha, Anis Zouaoui, Shahrear Iqbal, Xueyuan Han, Thomas Pasquier

Few Edges Are Enough: Few-Shot Network Attack Detection with Graph Neural Networks

IWSEC 2024 (AR: 29.8%), *best paper*

[[paper](#), [code](#), [slides](#)]

[Tristan Bilot](#), Nour El Madhoun, Khaldoun Al Agha, Anis Zouaoui

A Survey on Malware Detection with Graph Representation Learning

ACM Computing Surveys, 2024

[[paper](#)]

[Tristan Bilot](#), Nour El Madhoun, Khaldoun Al Agha, Anis Zouaoui

A Benchmark of Graph Augmentations for Contrastive Learning-Based Network Attack Detection with Graph Neural Networks

CSNet 2023

[[paper](#), [poster](#)]

[Tristan Bilot](#), Nour El Madhoun, Khaldoun Al Agha, Anis Zouaoui

Graph Neural Networks for Intrusion Detection: A Survey

IEEE Access, 2023

[[paper](#)]

[Tristan Bilot](#), Nour El Madhoun, Khaldoun Al Agha, Anis Zouaoui

PhishGNN: A Phishing Website Detection Framework using Graph Neural Networks

SECURITY 2022

[[paper](#), [code](#), [slides](#)]

[Tristan Bilot](#), Grégoire Geis, Badis Hammi

Publications (Under review)

Learning Transferable Representations from Operating System Entities via Provenance Graph Distillation

Under review, NeurIPS 2026

EV-AUDIT: A Co-Evolutionary Auditing Framework for Task Hijacking in Multi-Agent Systems

Under review, NeurIPS 2026

PERCEPTOR: Temporal Graph Transformer for Large-Margin Provenance-based Intrusion Detection

Under review, NDSS 2026

LARES: Targeted Lateral Movement Detection in Evolving Networks Through Source Host Identification

Under review, USENIX Sec 2026

Technical Articles

USENIX ;login: – Article based on our two USENIX Security 2025 papers

2025 [[article](#)]

Medium – Various articles on MLX/CUDA benchmarks, data eng., software eng.

2020–now [[11 articles](#) + [code](#)]

Personal Blog – “Deep learning from scratch” series, on autodiff & backpropagation

2022 [[5 articles](#), [code](#)]

Invited Talks

Ecole Polytechnique de Louvain – PIDSMaker Tutorial

Belgium – 2026 [[slides](#)]

NDSS'26 PRISM Workshop – PIDSMaker Tutorial

USA – 2026 [[slides](#), [video](#)]

Télécom Paris – Class on AI techniques for advanced attack detection

France – 2026 [[slides](#)]

The University of Texas at El Paso – Achieving High Quality of Attribution in IDS

USA – 2025 [[slides](#)]

GenAI Meetup Morocco – How AI protects us from cyberattacks?	Morocco – 2025 [slides]
EPITA Seminar – Introduction to ORTHRUS and PIDSMaker	France – 2025 [slides]
University of British Columbia – Inductive Host Detection in Large Temp. Graphs	Canada – 2024 [slides]
Institut Mines-Télécom – System-level IDS with Graph Neural Networks	France – 2024 [slides]
DATAIA Day Saclay – Detecting Complex Attacks with Graph Deep Learning	France – 2022 [poster]
Carrefour – Data Engineering Applied to Retail	France – 2022 [slides]

Projects

PIDSMaker – Deep learning framework for building provenance-based IDS	2025– [code]
Apple MLX – Implemented backpropagation of scattering operations in C++	2024– [code]
MLX-graphs – GNN library on top of MLX with optimized GPU kernels	2024– [code]
MLX-benchmark – Benchmark framework for MLX, Apple chips and CUDA GPUs	2024– [code]
Deepiler – Transformer-based decompiler to convert binaries into C code	2022 [code]
K – x86 Kernel – Simple kernel written in C and Assembly x86	2021 [code]

Program Committees

IEEE Symposium on Security and Privacy (IEEE S&P)	2027
Learning on Graphs (LoG)	2026
IEEE Transactions on Networking	2025
KES Detection of Complex Attacks	2025
Information Security Journal: A Global Perspective	2025

Skills

Programming: Python, Swift, C, C++, Bash, JS, Java, CUDA, Assembly x86, Rust
ML Frameworks: PyTorch, MLX, Jax/Haiku, pandas, scikit-learn
Agentic AI: Autogen/AG2, AWS Bedrock, LangChain, LangGraph
ML Skills: Self-supervised learning, GNNs, Transformers, LLM fine-tuning (LLaMA+QLoRA), GPU parallelization & vectorization, framework coding, large-scale training under limited resources
Infrastructure: GCP, AWS, Docker, Kubernetes, Airflow, dbt, W&B
Languages: French (native), English (fluent), Spanish (notions)

Activities & Interests

Volunteering: Protection civile (2018–2022), first aid
Academic: Student bureau, Junior Entreprise, class representative, hackathons (Google HashCode, Design4Green, Carrefour)
Hobbies: Music (DJ mix), cosmology, traveling, surfing